

2.2 DIAGNOSTIC AND TROUBLESHOOTING

Some people get a little excited when they look inside a computer and see all the different electrical components and circuit boards. All the wires, connectors, and data cables inside tend to be a little intimidating. Yet, all of today's computer repairs, replacement, upgrades and installations are getting easier and easier.

A technician could spend hours to search for a specific chip or failed solder connection that is causing a particular problem. Repairs are not done at the chip level anymore. Everything is very modular. It is quicker, easier, and much more economical to have the technician pop in a whole new video card or motherboard.

It is still important to know some of the different components and what they do. It can give an insight as to which particular module needs replacement. It might require a lot more time to troubleshoot the problem and discover which module needs replacement. This section serves as a guide on how to diagnose and troubleshoot desktop personal computers.

3.1 REPAIR AND MAINTENANCE TOOLS

Tools required for troubleshooting and repairing PC systems may be classified into basic, supplementary and specialized tools.

Basic Troubleshooting/Repair Tools:

The basic tools that should be in every troubleshooter's tool kit are as follows:

- Flat blade screwdrivers
- Philips screwdrivers
- Torx screwdrivers
- Nutdrivers
- Cleaning Swabs, canned air (just blown), and contact cleaner chemicals.

Supplementary Tool, Soldering/Repair Tools

Although the basic set of tools is useful, it should be supplemented with some other basic tools, such as:

- Electrostatic discharge (ESD) protection kit (includes wrist strap and mat)
- Needle-nose pliers and hemostats (curved and straight)
- Electric screwdriver
- Flashlight
- Wire cutter or stripper
- Vice or clamp
- Metal file
- Markers, pens, and notepads
- Spare CR-2032 lithium coin cell batteries
- Windows 98 startup floppy
- Windows 2000/XP (bootable) CD
- Diagnostic software
- Soldering and desoldering tools
- Nylon tie-ties
- Nylon tie-ties
- Digital multimeter

Specialized Troubleshooting/Repair Tools:

These advanced tools enable one to more accurately diagnose problems and make repairs faster. Jobs easier and faster.

- Memory-testing machines, used to evaluate the operation of SMMs, DIMMs, RIMMs, or CMMs
- DIMMs
- Serial and parallel loopback (or wrap) plugs to test serial and parallel ports

- Network cable scanner
- Serial breakout box
- A POST card (if you work with computers running DOS or other non-Windows operating systems, you might prefer to get a POST card that can also display IRQs and DMAs in use)
- Logic Probes and Logic Pulser
- Outlet Testers
- Temperature Probe or Infrared Thermometer

3.2 TROUBLESHOOTING TIPS AND TECHNIQUES

Troubleshooting PC hardware problems seems daunting, but in reality it is much simpler than it seems. Most problems can be diagnosed and corrected using few, if any, special tools and can be accomplished by anybody who can apply simple deductive reasoning and logical thinking. The bottom line with troubleshooting PC problems is that a solution exists, for every problem, and through simple practices combined with deductive reasoning, that solution can easily be found.

➤ Industry-Standard Replaceable Components

The use of industry-standard components is one of the key features of a PC. This means that virtually all the parts that make up a system are interchangeable with other systems in some manner. This also means that the parts are plentiful, inexpensive, and generally very easy to install. A typical PC contains the following replaceable components, most of which are made to industry standards for design and form factor:

- | | | |
|-----------------------------|-----------------|---------------------|
| ■ Motherboard | ■ Video card* | ■ CD-ROM/RW drive |
| ■ Processor | ■ Monitor | ■ DVD-ROM/+RW drive |
| ■ CPU heatsink/fan | ■ Sound card* | ■ Floppy drive |
| ■ RAM | ■ Speakers | ■ Drive cables |
| ■ CMOS battery | ■ Network card* | ■ Keyboard |
| ■ Chassis with optional fan | ■ Hard drive | ■ Mouse |
| ■ Power supply | | |

**May be integrated into the motherboard in some systems.*

Although some of the more well-optioned systems might have even more components than listed here, most PCs have fewer than 20 replaceable "parts." Some can have as few as 10-15, depending on how many options are present and how they are integrated. One of these components is either improperly installed (configured) or defective. If it is improperly installed or configured, the component can be repaired by merely reinstalling it or configuring it properly. If it's truly defective, the component must simply be replaced.

➤ Troubleshooting by Replacing Parts

There are several ways in which to troubleshoot PCs, but in the end it often comes down to simply reinstalling or replacing parts. In its simplest form, have two identical PCs sitting side by side. One of them has a hardware problem and the other works perfectly well. The simplest technique for finding the problem would be to swap parts from one system to another, one at a time, retesting after each swap. At the point when the faulty components are swapped, upon powering up and testing, the problem would be found to have moved from one system to the other. Knowing that the last item swapped just identified the source of the problem! Although this is very simplistic, it is often the quickest and easiest way to identify a problem component as opposed to specifically

testing each item with diagnostics. Instead of having an identical system standing by to borrow parts from, have an inventory of what is called "known-good spare" parts. These are parts that have been previously used, are known to be functional, and can be used to replace a suspicious part in a problem machine. However, this is different from new replacement parts which could be faulty. This technique is also effective because so few parts are needed to make up a PC and the known-good parts don't always have to be the same.

> Troubleshooting by the Bootstrap Approach

The "bootstrap approach" to PC troubleshooting is good for what seems to be a dead system. In this approach, the system is taken apart; strip it down to the chassis/power supply, bare motherboard, CPU (with heat-sink), one bank of RAM, and a video card with display and then power it up to see whether it works. In that stripped configuration, the POST or splash (logo) screen on the display should be seen, verifying that the motherboard, CPU, RAM, video card, and display are functional. If a keyboard is connected, the three LEDs (caps lock, scroll lock, and num lock) should flash within a few seconds after powering on. This indicates that the CPU and motherboard are functioning because the POST routines are testing the keyboard. After this, reinstall or add one part at a time, testing the system each time a change is made to verify it still works and that the part added or changed is not the cause of a problem. Essentially, it is rebuilding the system from scratch using the existing parts, but doing it one step at a time. Many times problems are caused by corrosion on contacts or connectors, so the mere act of disassembling and reassembling a PC will "magically" repair it. Although it might seem that nothing was changed and everything is installed exactly like it was before, in reality simply unplugging and replugging renews all the slot and cable connections between devices, which is often all the system needs.

> Useful Troubleshooting Tips

Some useful troubleshooting tips include:

- Eliminate unnecessary variables/components that are not pertinent to the problem.
- Reinstall, reconfigure, or replace only one component at a time.
- Test after each change you make.
- Keep a detailed record of each step taken (write it down).
- Do not give up! Every problem has a solution.
- If a roadblock is hit, take a break or work on another problem.
- Do not overlook the simple/obvious. Check the installation/configuration of each component.
- Keep in mind that the power supply is one of the most failure-prone parts in a PC. A high-output "known-good" spare power supply is highly recommended for testing.
- Cables and connections are also a major cause of problems, so keep replacements of all types on hand.

> Basic Troubleshooting Procedures

Before starting any system troubleshooting, a few basic steps should be performed to ensure a consistent starting point and to enable isolating the failed component:

1. Turn off the system and any peripheral devices.
2. Disconnect all external peripherals from the system, except for the keyboard and video display.
3. Make sure the system is plugged in to a properly grounded power outlet.

4. Make sure the keyboard and video displays are connected to the system.
5. Turn on the video display, and turn up the brightness and contrast controls to at least two-thirds of the maximum.
6. If there is no video display but the system seems to be working, try moving the card to a different slot or try a different video card or monitor.
7. To enable the system to boot from a hard disk, make sure no disk is in the drive. Or put a known-good bootable or diagnostics disk in the drive for testing.
8. Turn on the system. Observe the power supply/chassis fans, and lights on the system front panel. If the fans do not spin and the lights do not light, the power supply or motherboard might be defective.
9. Observe the power on self test (POST). If no errors are detected, the system beeps once and boots up.
10. Confirm that the operating system loads successfully.

➤ Problems During the POST

Problems that occur during the POST are usually caused by incorrect hardware configuration or installation. Actual hardware failure is a far less-frequent cause. If there is a POST error, check the following:

1. Are all cables correctly connected and secured?
2. Are the configuration settings correct in Setup for the devices installed?
3. In particular, ensure the processor, memory, and hard drive settings are correct.
4. Are all drivers properly installed?
5. Are switches and jumpers on the baseboard correct?
6. Are all resource settings on add-in boards and peripheral devices set so that no conflicts exist?
7. Is the power supply set to the proper input voltage (110V-120V or 220V-240V)?
8. Are adapter boards and disk drives installed correctly?
9. Is a keyboard attached?
10. Is a bootable hard disk installed or properly partitioned and formatted?
11. Does the BIOS support the drive installed? If so, are the right parameters entered?
12. Are all memory SIMMs or DIMMs installed correctly? Try reseating them.
13. Is the operating system properly installed?

➤ Hardware Problems After Booting

If problems occur after the system has been running, and without having made any hardware or software changes, a hardware fault possibly has occurred. Here is a list of items to check in that case:

1. Try reinstalling the software that has crashed or refuses to run.
2. Try clearing CMOS RAM and running Setup.
3. Check for loose cables, a marginal power supply, or other random component failures.
4. A transient voltage spike, power outage, or brownout might have occurred. Symptoms of voltage spikes include a flickering video display, unexpected system reboots, and the system not responding to user commands. Reload the software and try again.
5. Try reseating the memory modules (SIMMs, DIMMs, or RIMMs).

> Problems Running Software

Problems running application software (especially new software) are usually caused by or related to the software itself, or are due to the fact that the software is incompatible with the system. Here is a list of items to check in that case:

1. Does the system meet the minimum hardware requirements for the software?
2. Check to see that the software is correctly installed. Reinstall if necessary.
3. Check to see that the latest drivers are installed.
4. Scan the system for viruses using the latest antivirus software.

3.3 TROUBLE-SHOOTING COMMON MODULES:

Here are some tips on how to go about trouble-shooting common modules. Observe the symptoms, do the checks, and perform the actions in the order they appear and the problem may be solved before the last check. Remember to unplug the power supply from the wall and observe all ESD precautions before working inside the case.

> Power Supply Diagnostic Chart:

No	Symptoms	Checks	Action
1.	Power supply appears Dead. Screen appears dead. Fan - not running.	Check power source Check power supply cable Check 110/220v select switch Is motherboard lead installed? Check power and reset switches Is power connections to motherboard correct?	Repair/change as applicable. Replace if bad Select appropriate voltage Connect front panel switch lead Replace where applicable If yes see 3. if not correct the connections
2.	Power supply comes on. Fan is running. System does not boot.	Check if system boots on second try Listen if there are any beeps Is any new hardware installed?	Premature power-ok signal. Try a different power supply. Proceed to motherboard trouble shooting Remove latest addition and retry. Test the hardware removed. Replace power supply
3.	Power supply comes on. System does not boot. No new hardware installed.	If hard drive does not spin, check if it spins on other leads If hard drive spins, check for bad adaptors on bus. Check if the supply powers the motherboard on bench.	If yes, defective power Supply lead or connector. If not, try drive in test PC Replace faulty adaptor card. If not replace power supply

> Trouble shooting the video card

No	Systems	Check	Action
1	System powers up. No live screen	Check monitor power source Check monitor power supply cable Check monitor 110/220V select switch Check video control positions Is there any string of beeps?	Repair/change as application Replace if bad Select appropriate voltage Adjust brightness and contrast See 2

2	System powers up Screen alive. Only a blinking cursor is displayed	Is there any string of beeps?	Reseat video adapter If there is no beep see 3
		Check if RAM is well seated.	Reseat RAM
		Check for damaged cable or bent pins	See 3
		Check video cable	Connect video cable
3	System powers up Screen alive. No blinking cursor	Check for damaged cable or bent pins	Répair or replace video cable
		Check if other adapters are installed	Remove the other adapters and retry. Replace adapter if no other adapters are installed. If the problem persist go to motherboard trouble shooting
4	System powers up Screen alive. Freezes on BIOS screen	Check other adapters are installed	See 3

- Motherboard diagnostic chart

No	Systems	Check	Action
1	System power up No live screen Power diagnostic done Video diagnostic done	Check if RAM modules are seated	Reseat RAM modules
		Check if RAM modules matched.	Install pairs if required.
		Check if CPU is seated.	Reseat CPU and fan.
		Check if fan on CPU is active	Replace fan or its power point as applicable
		Are there any beeps?	Replace RAM
		Check motherboard settings	Restore motherboard setting
2	System power up Screen alive. Freezes on boot Screen. Freezes bare bones	Check if motherboard runs on bench	See 3
		Check whether system freezes on swapped RAM?	If not replace RAM.
		Check CMOS setting	Load defaults from BIOS
		Check if heat sink is active.	If not install new heat sink
		Check if motherboard runs on bench	Locate short or swap case
		Check if system works on swapped CPU	If yes replace CPU If not replace motherboard.

> Hard Drive Diagnostic

No	systems	check	Action
1.	System powers up Screen alive. BIOS doesn't register drive	Check master/slave setting on drive	Set switches as appropriate
		Check cable select (CS)	Select master/slave or CS
		Check if cable are well seated	Reseat IDE and power cables
		Check if drive spins on power up	Try swapping cables and test drive before trashing
		Check if IDE ribbon cable is Keyed	Replace IDE cables or match pin 1 on all connectors as applicable
		Check if BIOS now register drive.	If not either motherboard controller or drive is bad or incompatible

2	System powers up Screen alive BIOS register drive	Check if drive cycles up and down	Swap power lead and try isolating on IDE cable
		Check transfer speed mode in CMOS.	Set as appropriate
		Check if FDISK see drive	If not isolate drive on Primary IDE controller
		Check if FDISK still doesn't see drive	Try FDISK/MBR

> CD/DVD Trouble shooting

NO	SYMPTOMS	CHECKS	ACTION
1.	Drive try won't eject.	Check if BIOS doesn't Register drive	Proceed as in hard drive failure
2.	Drive tray ejects Drive doesn't read data disc	Does drive have a pinhole on its face?	If yes insert pin and pull tray.
		Check if the right drive is being read	If not remove face plate and pry
		Check if drive is not installed	Pick the right drive
		Check if drive read other discs.	Install drive
		Check if device manager has problem	Clean disc, check for scratches
3.	Drive tray ejects Drive reads data disc	Check if drive shares IDE cable	Reinstall or update drive OS
		Check if drive doesn't play music CDs	Isolate drive on secondary IDE
		Check if drive vibrates or is noisy	Replace IDE cable. Replace drive
		Check if drive doesn't boot CD	Replace audio patch cable
			Eject and remount. Try another disc. Make sure drive is leveled.
			Change CMOS boot sequence
			Pre-spin drive. Replace drive.

> Sound Card Trouble shooting

No	Symptoms	check	Action
1.	System power up No sound	Check if sound is not enabled in OS	Enable sound and test clips
		Are external speakers not powered?	Plug in speaker bricks
		Are external volume controls not up.?	Turn up external volume controls
		Check if the best drive is not loaded	Download latest drive from internet
2.	System power up. No sound Device manager register drive	Check for "I" or "J" in device manager	Proceed to conflict resolution
		Check all jacks	Connect speaker jack to correct port
		Check every software volume mute	Un-mute in each software volume
		Is problem with music CDs only?	Replace patch cables. Replace speaker. Replace sound card
3.	System power up. No sound. Device manager does not register drive	Does device manager register sound card with only video card installed?	Proceed to conflict resolution
		Is the sound card build-in on the motherboard?	If yes disable on-board sound in CMOS and install new sound card. If not replace sound card.

3.4 REPLACING COMMON MODULES

Since repairs are not at the chip level anymore, after diagnosing the faulty module, it is replaced with a new functional module. Here are the procedures for the replacement of some common modules.

> REPLACING THE SUPPLY:

For people who have never worked on a PC before, the power supply is one of the toughest components to replace simply because of the large number of connections. Here is the procedure.

- Unplug the power cable from the back of the power supply
- Remove the cover from the case. On a standard mid-tower case, only a side cover is removed (two screws from the back and it slides right out)
- Remove the four mounting screws (at the back of the computer) that hold the power supply in place
- Move the power supply out of the way on top of the drive cage leaving it connected to serve as a guide
- Put the new power supply in place and install the four screws to secure it.
- Replace the old power supply connectors with connectors from the new power supply
- Tie off extra leads with a wire tie to keep them from flopping all over the place
- Slide in and secure the case lid.

> REPLACING SOUND CARD

Sound card do not have a particularly high failure rate, but they get replaced more often than any other adapter, with the possible exception of modems

- Unplug the PC and open the case
- Remove the single screw that secured the old sound card case
- Disconnect the analog audio lead connecting the CD drive to the case
- Remove the old sound card by pulling it straight up and out of the case
- Install the new sound card by seating it in the PCI slot with even pressure on the bracket and its back edge
- Immediately after installing the sound card, secure it with screw through the bracket on the back rail
- Attach the CD audio lead to the sound card. Obviously, the other end is attached to the CD/DVD drive
- Cover up the case and install the driver that came with the adapter card

> REPLACING A VIDEO CARD

Video cards are pretty reliable, though some of them run so hot that they have their own cooling system, and if the fan fails, the video processor can't be too far behind. Use the following procedure to a replace video card

- Open up the case by removing the two screws at the back that secure the side cover and sliding it out
- Remove the video card hold-down screw
- Gently pull out the old video card from the slot
- Handle the new video card by the corners, not touching the gold contact fingers, and position it properly over the slot

- Seat the new video card evenly by pressing down on the top edge at both the front and back of the card
- Install the retaining screw on the back rail and inspect the card to make sure it is still seated in the slot
- Slide in and secure the case lid then install the software drivers that came with the video card.

➤ REPLACING THE HARD DRIVE

When a hard drive needs to be replaced, back up all the data that could still be accessed on floppy, then remove the old hard drive. Use the following procedure to replace.

- Open up the case by removing the two screws at the back that hold the side cover in place and sliding it out
- Remove the four screws that secure the hard drive to the cage. If need be remove the second side cover.
- Clear a path and slide the hard drive out.
- Remove the broad ribbon cable which carries the data and the 4x1 power cable.
- Set the jumper setting for the Master/slave selection on the new drive the same way as in old drive
- Install the IDE ribbon cable and the power on the replacement hard drive while outside the case.
- Clear plenty of room behind the cage and slide the hard drive straight back into place then secure it
- Slide back and secure the side cover(s)

➤ REPLACING CD/DVD DRIVE

CD/DVD drives are delicate and can vibrate like crazy if they are not secured properly. The following steps show how to go about replacing them

- Remove both sides of the case
- Remove the four screws (two on each side) that secure the drive to the bay
- Unplug the data power and audio patch cables from the back of the CD/DVD drive
- Slide out the drive from the front of the drive bay.
- Put the new drive next to the old drive and select the same jumper pair for master/slave setting.
- Install the new drive into the case, and secured with four screws. Make sure the face of the drive flush with the front facade before tightening the four screws.
- Reconnect the audio patch, data and power cables. Where the data is not keyed, match the red wire on one side of the flat ribbon to the "Pin 1" end of the connector on the CD drive
- Slide in and secure the side covers.

➤ REPLACING RAM:

There are two good reasons for installing RAM in a system, either for upgrading RAM, or the current module has failed. It is a simple job and here is how to go about it.

- Open up the case by removing the two screws at the back that secure the cover and sliding it out.

- Remove the old RAM carefully using both thumbs (or two fingers) to simultaneously depress the white locking levers on the memory slot.
- Check the labeling to make sure the replacement RAM will work.
- Hold the replacement RAM module by the edges and align the notches with slots before installing it.
- Seat the memory module by pressing down firmly with your thumbs on both ends of the module. The white locking ears will rise into place of their own accord if the RAM is installed properly.
- Slide in and secure the case lid

> REPLACING THE CPU:

Despite the high clock speeds and high heat dissipation of modern CPU, their failure is rare. Replacing a CPU is not particularly challenging, but it can be a little intimidating for the first timer. Here is how:

- Open the case.
- Disconnect the power to the heatsink fan.
- Remove the heatsink by studying the latching mechanism then use the thumb to release it.
- Release the locking lever by lifting it up to the vertical position.
- Remove the CPU by lifting it straight out of the socket.
- Insert the new CPU with the key at the edges aligned.
- Lock the CPU into place with the locking lever, which is pulled a little away from the side of the socket to get past the locking nub.
- Apply little (about the size of a large grain of rice) of thermal compound (ceramique) on the CPU.
- Install the heatsink by first latching the fan side of the retaining spring and then latching the rear side.
- Reconnect the power to the heatsink fan to the CPU fan point on the motherboard.
- Replace the case cover.

> REPLACING THE MOTHERBOARD:

The first step to replacing a motherboard in a desktop computer is to remove the old motherboard. That may sound trivial, but it's literally half of the job. Hence this section is splitted into removal and installation

Removal:

- Remove the cover from the case
- Unplug the 1x4 power connectors from the hard drive.
- Remove PCI adapters and the video card by first removing the securing screws on the back rail and pulling out each one from its slot.
- Pull out the 20 pin power supply connector from the motherboard by first depressing the top of the latch.
- Remove the data cables from the old motherboard. Use both hands and hold the ribbon cable as near to the connector as possible to pull it out. For a roomy ccse, leave them attached to the drives on the other end.
- Remove the series of screws holding the motherboard to the case.
- Disconnect the front panel leads that attach to the motherboard. This includes the LED for hard drive activity and power status, the case speaker, and most importantly the power switch.

- Remove the motherboard by raising it a little and pulling it away from the back of the case, and then lifting it right out of the case.

Installation:

- Compare the old motherboard with the replacement motherboard to see how they physically match
- Compare the locations of the holes in the motherboard with the standoffs in the case.
- Install the motherboard into the case on an angle, with the back edge and I/O core going first.
- With the I/O core properly aligned with the shield, push the ports through the opening and set the motherboard down on the standoffs.
- Check for wires caught below the motherboard before securing it with screws.
- Connect the power switch and the rest of the front panel leads to the motherboard.
- Load the adapters and the video card back in and secure each with a hold down screw on the back rail. Make sure the adapters are seated well in the motherboard
- Reconnect the 20 pin ATX power connector to the motherboard
- Reconnecting all the drive cables, power and data that was undone when removing the old motherboard
- Tie up long cables to neaten up the case a little and encourage better air circulation.
- Close up the case.

4.0 SAFETY AND PREVENTIVE MAINTENANCE PROCEDURES ✓

This section identifies safety procedures that should be followed when working on a computer system. It also identifies ESD precautions and procedures and states the steps and procedures for preventive maintenance.

4.1 SAFETY PROCEDURES

Before working on a system, there are certain safety procedures that should be followed. Some are to protect the person working on the system, whereas others are to protect the system. Here are some:

- Unplug a PC from the wall before working on it.
- Be careful when working on power supplies as most of them have 400 volts present at some points internally.
- Colour displays have between 50,000 and 100,000 volts on the CRT! Be sure to discharge the tube (line transformer) before working on it.
- Be extra careful when handling the vacuum tube itself as it can implode if broken or punctured, posing high level of risk.
- While working on a PC, wear an ESD wrist strap that is clipped to the chassis of the machine. This prevents static electricity from damaging the system as it is touched.
- Use commercially available wrist strap that feature an internal 1-mega-ohm resistor designed for protection, rather than making one.
- Components removed from the system should be placed on a special conductive antistatic mat or bag, which is also a part of any good ESD protection kit.
- Devices sealed with tamperproof Torx screws generally have high voltages or other dangers waiting inside, so being able to bypass these screws does not mean it is always a good idea.
- Some screws in a system can be length-critical, especially screws used to retain hard disk drives. Some hard disks are destroyed by using longer mounting screws.
- When using a multimeter to test voltages that could potentially be 50V or above (such as AC wall socket voltage), always use one hand to do the testing, not two.
- Be aware that many analog meters can be dangerous to digital circuits. These meters use a 9V battery to power the meter for resistance measurements.
- The use a surge protector will not protect the system from an improperly wired outlet; therefore, use an outlet tester to ensure the outlet is computer friendly.
- Note that using an electric screwdriver when installing a motherboard can be dangerous because the bit can easily slip off the screw and damage the board.

4.2 ELECTROSTATIC DISCHARGE

Electrostatic discharge (ESD) is the sudden and momentary electric current that flows between two objects at different electrical potentials. The term is usually used in the electronics and other industries to describe momentary unwanted currents that may cause damage to electronic equipment.

➤ Causes of ESD

One of the causes of ESD events is static electricity. Static electricity is often generated through tribocharging, the separation of electric charges that occurs when two materials are brought into contact and then separated. The friction between the two materials results in tribocharging, thus creating a difference of electrical potential that can lead to an ESD event.

Another cause of ESD damage is through electrostatic induction. This occurs when an electrically charged object is placed near a conductive object isolated from ground. The presence of the charged object creates an electrostatic field that causes electrical charges on the surface of the other object to redistribute and make it have regions of excess positive and negative charges. An ESD event may occur when the object comes into contact with a conductive path.

The most spectacular form of ESD is the spark, which occurs when a strong electric field creates an ionized conductive channel in air. This can cause minor discomfort to people, severe damage to electronic equipment, and fires and explosions if the air contains combustible gases or particles.

However, many ESD events occur without a visible or audible spark. A person carrying a relatively small electric charge may not feel a discharge that is sufficient to damage sensitive electronic components. Some devices may be damaged by discharges as small as 12 volts. These invisible forms of ESD can cause device outright failures, or less obvious forms of degradation that may affect the long term reliability and performance of electronic devices. The degradation in some devices may not become evident until well into the service life of some devices.

➤ ESD Prevention

Here is what needs to be done to reduce the risk of ESD damage:

- Wear a ground bracelet
- As components are removed, store them in anti-static bags
- Use a grounded static mat underneath the computer
- Remove all jewelry
- Those that have long hair should tie them back
- Wear natural rather than synthetic clothing
- Work on hard floors rather than carpeted surfaces
- Raise the humidity of the working environment

4.3 PREVENTIVE MAINTENANCE

Much as the name implies, preventive maintenance, refers to performing proactive maintenance in order to prevent system problems. This section discusses some of the general concepts regarding preventive maintenance, the different types that are relevant to PCs, and how to set up a preventive maintenance schedule.

➤ Importance of Preventive Maintenance

Preventive maintenance is one of the most ignored aspects of PC ownership. Most people seem to think that the PC does not need preventive maintenance. These people generally find themselves repairing or replacing much sooner than those that take definite steps to avoid difficulties in the first place. Here are some reasons for developing a preventive maintenance plan for the PC:

- Preventive maintenance saves money.
- Preventive maintenance saves time.
- Preventive maintenance helps safeguard data.
- Preventive maintenance improves performance.
- Preventive maintenance reduces component failure.
- Preventive maintenance ensures long life for the system.
- Preventive maintenance increases a system's resale value.

➤ Determining When to Perform Maintenance

Some types of preventive maintenance need to be performed more often than others. The frequency of preventive maintenance depends on the nature of the activity; some things just need to be addressed more often than others. It also depends a lot on what the PC is being used for.

How often one should perform active preventive maintenance procedures depends on the system's environment and the quality of the system's components. If the system is in a dirty environment, such as a machine shop floor or a gas station service area, the system might need cleaning every three months or less. For normal office environments, cleaning a system every few months to a year is usually fine. However, if the system is opened after one year and dust bunnies are found inside, the cleaning interval should probably be shortened.

One enemy of preventive maintenance is simply remembering to do it. One way to address this problem is through the use of a preventive maintenance schedule, which helps remind one of when to perform key maintenance activities on the PC.

Developing a preventive maintenance program is important to everyone who uses or manages personal computer systems. The two types of preventive maintenance procedures are active and passive.

➤ Active Preventive Maintenance Procedures

An active preventive maintenance program includes procedures that promote a longer, trouble-free life for your PC. This type of preventive maintenance primarily involves the periodic cleaning of the system and its components. The following sections describe several active preventive maintenance procedures.

Cleaning the monitor (every 2 – 6 months)

Flat-screen LCD monitors and chunkier CRT displays must be cleaned in different ways. This is how to go about cleaning CRT displays;

- Switch off and unplug the screen before cleaning it.
- Leave the display switched off overnight before cleaning it.
- Squirt a standard glass cleaner on a lint free cloth.
- Use the dampened cloth to wipe the glass.

For LCD screens, the procedure is as follows;

- Switch off and unplug the screen before cleaning it.
- Use a soft lint free cloth dampened with a little water.
- Wipe very gently and do not press into the display.

Cleaning the keyboard (every 2 – 6 months)

There are several ways to clean a keyboard. The easiest is to hold the keyboard upside down over a dustbin and tap the bottom to knock out any crumbs or dust. For a more thorough clean, proceed as follows;

- Get a can of compressed air and spray it into the cracks in the keyboard.
- Run a moistened cotton bud through the keys to clean any dirt stuck between them.
- Wipe the entire keyboard with an antibacterial wipe to remove any nasties that may lurk there.
- If this still is not enough, write the sequence of the keys down or take its digital photograph.
- Ease a screwdriver or blunt knife under each key and gently lift the key top off.

- Once the keys are off, use a can of compressed air to remove debris from inside the keyboard.
- Clean the keys using antibacterial wipes or wash them.
- Leave the keys to completely dry.
- Press each key back into the keyboard gently until a click confirms that it is secured.

Cleaning the mouse (every 2 – 6 months)

Just like keyboards, mice can pick up bits of dust and dirt, and this can eventually stop them from working properly. To clean a mouse,

- Disconnect the mouse from the computer
- Spray isopropyl rubbing alcohol onto a lint-free cloth.
- Use the cloth to remove dirt from the body of the mouse without damaging the plastic.
- Pay extra attention to the left and right click buttons.
- Turn the mouse on its back.
- If this shows the top of a ball, look for a disc that holds the ball in place.
- Turn the disk in the direction indicated by the arrows and remove both the disc and the ball.
- If the rotating wheels inside the mouse are dirty, remove the grime using a tweezer.
- Put the mouse back together and reconnect it to the back of the computer.
- If the mouse does not have a ball, do not clean the bottom of the mouse with a cloth.
- Look for a few raised pads on the base of the mouse.
- Any dirt on these can be carefully scraped away with a small knife or screwdriver.

Cleaning the case (every 2 – 6 months)

Once the peripherals have been taken care of, it is time to step up the cleaning to the computer itself. Here is how to go about it.

- Unplug the PC from the mains and turn off the switch on the power supply if there is one.
- Remove the PC's side panel.
- Set the open PC in a well-lit working area.
- Wear an anti-static strap connecting it to a grounded metal surface of the system.
- Get a can of compressed air, a blower or a vacuum.
- Spray onto the dusty parts so that the particles exit from the back of the PC.
- Keep the nozzle a decent distance from any components when doing so.
- If any large fan blades are accessible, carefully wipe them with a lint-free cloth to remove dust.
- The plastic case can be cleaned with a lint-free cloth that has been slightly dampened with water.
- Verify that all items such as RAM, video cards and NICs are still securely seated.
- Check all the cables and plugs inside the PC and make sure they are fastened securely.
- Finally close the PC case.

Tips on keeping a computer clean

Once the computer is clean it is relatively easy to keep it free from dirt. Simply follow these tips:

- Place the computer in a well-ventilated room
- Close notebook computers when not in use
- Place a plastic cover over the monitor when not in use
- Always use a mouse mat

- Never eat or drink near the computer or over the keyboard
- Do not smoke near the computer or in the room where it is placed

Monitor disk space (every week)

Disk space, in particular 'free disk space', is important to any operating system that is installed on a PC. The operating system (such as Microsoft Windows) needs a certain amount of free disk space for swap. Always keep a minimum of 5-10% of the disk free and a manual check for free disk space is adequate. It is done as follows:

- Click 'Start'
- Click 'My Computer'
- Right click C:\ drive.
- Click 'Properties'
- A simple pie chart will be displayed that shows free disk space.

Monitor temporary files (monthly)

The functionality of operating systems such as Microsoft Windows relies on temporary files and temporary Internet files created by applications (such as word processors) and Internet browsers. The TEMP folder is different for each user. On a Windows 2000 Professional PC the folder path will be something like this:

C:\Documents and Settings\tsmith\Local Settings\Temp (where tsmith is the username)

If unsure of the path simply open a Command window and type the command 'set' and press enter. Within the list displayed will be a TEMP setting with the exact path.

The temporary files created by the Internet browser (such as Internet Explorer) are kept in a different folder and are managed a little different. The typical path for the temporary Internet files is:

C:\Documents and Settings\tsmith\Local Settings\Temporary Internet Files (where tsmith is the username)

Managing and deleting these files is best done through Internet Explorer. Open Internet Explorer and click Tools > Internet Options from the menu. Use the [Delete Files] button to remove the existing files. Click the [Settings] button to control how Internet Explorer manages the temporary Internet files.

The most valuable option here is the ability to allocate a specific amount of disk space to the storage of temporary Internet files. Setting this to a low value like 20MB will help manage disk space and performance.

Monitor anti-virus definition files (weekly)

With new viruses being introduced almost daily, it has never been more important to keep your virus detection software up to date. The industry leading anti-virus software producers build auto-update features into their products to make updating very convenient but it is always a good idea to double check that the autoupdate process is doing the required job. From the System Tray (icons at bottom right near clock) right click the V-Shield icon and select 'About' from the pop-up menu to display the information about the associated definition files. If the definition files are over 1 week old manually update them as soon as possible.

Hard disk maintenance (monthly)

For Microsoft Windows users there are a couple of Disk Maintenance items that are rarely completed (Scandisk and Defragment). In most cases these tasks are known to the user but simply forgotten. To run a scandisk follow this procedure:

- Click Start
- Click My Computer
- Right click C:\ drive
- Select Properties from the pop-up menu
- Click tools to access the Disk Management tool screen
- To run a scandisk click the [Check Now] button.
- Select both the Automatically Fix and Bad Sectors options
- Click Start.

A message stating that the scandisk program need exclusive access to the disk and offering to do a scandisk at the next reboot will be displayed.

To access the Defragmentation proceed as follows:

- Click Start
- Click My Computer
- Right click C:\ drive
- Select Properties from the pop-up menu
- Click tools to access the Disk Management tool screen
- Access the Defragmentation tool by clicking on the [Defragment Now] button
- To analyze the disk before performing a defrag, click the [Analyze] button.
- When you can live without your PC for a while click the [Defragment] button to start the process.

A colour based chart will show you the amount of fragmentation. The more red in the chart you see the worse the defragmentation is.

> Passive Preventive Maintenance Procedures

Passive preventive maintenance includes steps to take to protect a system from the environment, such as using power-protection devices; ensuring a clean, temperature-controlled environment; and preventing excessive vibration. In other words, passive preventive maintenance means treating the system well. Passive preventive maintenance involves taking care of the system by providing the best possible environment—both physical and electrical—for the system. Physical concerns are conditions such as ambient temperature, thermal stress from power cycling, dust and smoke contamination, and disturbances such as shock and vibration. Electrical concerns are items such as ESD, power-line noise, and radio-frequency interference.

Oddly enough, one of the most overlooked aspects of microcomputer preventive maintenance is protecting the hardware—and the sizable financial investment it represents—from environmental abuse. Before you set up a new PC, watch the following

- Place the PC in a place that is free of airborne contaminants such as smoke or other pollution.
- Do not place your system in front of a window
- The environmental temperature should be as constant as possible
- Provided power to the PC through properly grounded outlets
- Power should be stable and free from electrical noise
- The computer system should be on its own circuit with its own circuit breaker

- Avoid extension cords unless absolutely necessary, and use only heavy-duty extension cords.
- Keep the system away from radio transmitters or other sources of radio frequency energy
- Do not use computer desks that place the system unit in a sealed cabinet.

Observe the following after the PC is installed and commissioned

- If the system is not left on continuously, give it 15 minutes or more to warm up after a cold start before writing to the hard disk.
- If the system left on for long periods of time, make sure the screen is blank or displays a random image if the system is not in use.

5.0 BIOS BASICS

BIOS stands for Basic Input/Output System, which consists of low-level software that controls the system hardware and acts as an interface between the operating system and the hardware. It is essentially the link between hardware and software in a system.

The BIOS consists of all the individual driver programs that operate between the operating system and the actual hardware. As such, the operating system never communicates with the hardware directly; instead, it must always go through the appropriate drivers. It is usually the responsibility of the hardware manufacturer to provide drivers for its hardware. The drivers typically are operating system specific and consist of software that acts to interface the hardware to the software. The BIOS adapts to the unique hardware on one end but looks consistently the same to the operating system at the other end. The hardware is where most differences lie between various systems. It is up to the BIOS to mask the differences between unique hardware so that the given operating system (and subsequently the application) can be run.

5.1 BIOS HARDWARE/SOFTWARE

The BIOS itself is software running in memory that consists of all the various drivers that interface the hardware to the operating system. The BIOS is unique compared to normal software in that it does not all load from disk; some of it is preloaded into memory chips (ROM) installed in the system or on adapter cards. The BIOS in a PC comes from three possible sources:

- Motherboard ROM
- Adapter card ROM (such as that found on a video card)
- Loaded into RAM from disk (device drivers)

➤ The Motherboard ROM

This is most often associated with hardware rather than software. This is because the BIOS on the motherboard is contained in a ROM chip on the board, which contains the initial software drivers needed to get the system running. The motherboard BIOS usually includes drivers for all the basic system components, including the keyboard, floppy drive, hard drive, serial and parallel ports, and more.

➤ Adapter Card ROM

Because some drivers must be active during boot time and are not available on the motherboard, the PC's motherboard ROM is programmed to scan the slots looking for adapter cards with ROMs on them. If a card was found with a ROM on it, the ROM was executed during the initial system startup phase, before the system begins loading the operating system from the hard disk. They include video cards, SCSI adapters, network cards, ATA or floppy upgrade boards and V2K boards.

➤ Loaded into RAM from Disk

New hardware for which no motherboard BIOS drivers existed and do not have to be active during boot time such as CD-ROM drives, SCSI hard disks, USB ports and so on, are copied to the system hard disk and configured for the operating system to load them after boot time. This is far simpler and more practical than requiring a new motherboard BIOS that would specifically support the new devices.

5.2 MOTHERBOARD BIOS

All motherboards have a special chip containing software called the ROM BIOS. This ROM chip contains the startup programs and drivers used to get the system running and act as the interface to the basic hardware in the system. Simply put, the BIOS in most PCs have four main functions:

- **POST (Power On Self Test):** The POST tests the computer's processor, memory, chipset, video adapter, disk controllers, disk drives, keyboard, and other crucial components.
- **Setup:** This menu-driven program, activated by pressing a special key during the POST, enables one to configure the motherboard, chipset, date and time, passwords, disk drives, and other basic system settings. BIOS Setup also allows one to control the power-management settings, boot-drive sequence, and to configure CPU timing and clock-multiplier settings on some systems.
- **Bootstrap loader:** A routine that reads the first physical sector of various disk drives looking for a valid Master Boot Record (MBR). If one is found, the code within is executed and then continues the boot process by reading the first physical sector of the bootable volume, and then loads the first operating system startup file, upon which the operating system is then in control and continues the boot process.
- **BIOS (Basic Input/Output System):** This refers to the collection of actual drivers used to act as a basic interface between the operating system and the hardware when the system is booted and running. When running DOS or Windows in safe mode, you are running almost solely on ROMbased BIOS drivers because none are loaded from disk.

➤ ROM Hardware

The main ROM BIOS is contained in a ROM chip on the motherboard, but adapter cards with ROMs contain auxiliary BIOS routines and drivers needed by the particular card, especially for those cards that must be active early in the boot process, such as video cards. Cards that don't need drivers active during boot (such as sound cards) typically don't have a ROM because those drivers can be loaded from the hard disk later in the boot process. Because the BIOS is the main portion of the code stored in ROM, we often call the ROM the ROM BIOS.

5.3 CMOS SETUP

The CMOS RAM must be configured with information about your system's drives and user-selected options before the computer can be used. The Setup program provided with the system is used to select the options required to start the computer.

➤ Running or Accessing the CMOS Setup Program

To run the BIOS Setup program, reboot and press a particular key or key combination during the POST. Normally, the correct key to press is displayed onscreen momentarily during the POST. If it flashes past too quickly to read, try pressing the Pause key on the keyboard when it appears. This freezes the system, allowing you to read the display. Pressing any other key un-pauses the system and allows the POST to continue. The major vendors have standardized the following keystrokes to enter the BIOS Setup in recent systems:

- For *AMI BIOS*, Press Delete during POST.
- For *Phoenix BIOS (FirstBIOS Pro)*, Press F2 during POST.
- For *Award BIOS (FirstBIOS)*, Press Delete or Ctrl+Alt+Esc during POST.
- For *Microid Research (MR) BIOS*, Press Esc during POST.

At the BIOS Setup main screen, there is usually a main menu allowing access to other menus and submenus of different sections or screens. Though most motherboards feature slightly different settings, most are very similar.

> BIOS Setup Menus

Most modern BIOSs offer a menu bar that controls navigation through the various primary menus at the top of the screen when in the BIOS Setup. A typical menu bar offers the choices shown in Table 5.1.

Table 5.1: Typical BIOS Setup Menus

Setup Menu Screen	Description
Maintenance	Specifies the processor speed and clears the setup passwords. This menu is available only in Configure mode, set by a jumper on the board.
Main	Allocates resources for hardware components.
Advanced	Specifies advanced features available through the chipset.
Security	Specifies passwords and security features.
Power	Specifies power management features.
Boot	Specifies boot options and power supply controls.
Exit	Saves or discards changes to the setup program options.

Maintenance Menu

The Maintenance menu is a special menu for setting the processor speed and clearing the setup passwords. Older motherboards used jumpers to configure the processor bus speed (motherboard speed) and processor multiplier. Most new boards now offer this control via the BIOS Setup and only one jumper, called the configuration jumper, still remains on the board. Setup displays this menu only if the system is set in Configure mode. The Maintenance menu is as shown in Table 5.2.

Table 5.2: Typical Maintenance Menu Settings

Feature	Options	Description
Clear All Passwords	OK (default) Cancel	Clears the user and supervisor passwords
Clear BIOS	OK (default) Cancel	Clears the Wired for Management Boot Integrity Service (BIS) credentials
CPU Stepping Signature	No options	Displays CPU's stepping revision
CPU Microcode Update Revision	No options	Displays CPU's microcode update revision

Main Menu

The main menu in a modern system reports system information such as the BIOS version, the processor type and speed, the amount of memory, and whether the memory or cache is configured for ECC functionality. The main menu also can be used to set the system date and time. Table 5.3 shows a typical main menu.

Table 5.3: Typical Main Menu Settings

Feature	Options	Description
BIOS Version	—	Displays the version of the BIOS.
Processor Type	—	Displays the processor type.
Hyper-Threading Technology	Enabled (default); Disabled	Enables/disables Hyper-Threading Technology. This option is present only when a processor that supports Hyper-Threading Technology is installed.
Processor Speed	—	Displays the processor speed.
System Bus Speed	—	Displays the system bus speed.
System Memory Speed	—	Displays the system memory speed.
Cache RAM	—	Displays the size of the L2 cache
Total Memory	—	Displays the total amount of RAM.
Memory Mode	—	Displays the memory mode (dual channel or single channel).
Memory Channel A Slot 0; Memory Channel A Slot 1; Memory Channel B Slot 0; Memory Channel B Slot 1	—	Displays the amount and type of RAM in the DIMM sockets.
Memory Configuration	Non-ECC (default); ECC	ECC turns on error reporting if the system and all the memory installed supports ECC. This setting is available only when ECC memory is installed.
Language	English (default); others might be listed	Selects the current default language used by the BIOS.
System Time	Hour:minute:second	Specifies the current time.
System Date	Month/day/year	Specifies the current time.

Advanced Menu

The Advanced menu is a doorway to submenus for setting advanced features that are available through the motherboard chipset. This part of the BIOS setup is specific to the particular chipset the motherboard uses. Table 5.11 shows the typical Advanced submenus available.

Table 5.4: Typical Advanced BIOS Submenus

Submenu	Description
PCI Configuration	Configures the IRQ priority of individual PCI slots
Boot Configuration	Configures Plug and Play, resets PnP configuration data, and configures the NumLock key
Peripheral Configuration	Configures peripheral ports and devices
Drive Configuration	Configures ATA devices
Floppy Configuration	Configures the floppy drive

Event Log Configuration	Configures event logging
Video Configuration	Configures video features
USB Configuration	Configures USB support
Chipset Configuration	Configures advanced chipset features
Fan Control Configuration	Configures fan operation
Hardware Monitoring	Monitors system temperatures, voltages, and fan speeds

Security Menu

Most BIOSs include two passwords for security: the supervisor and user passwords. These passwords help control who is allowed to access the BIOS Setup program and who is allowed to boot the computer. The supervisor password is also called a setup password because it controls access to the setup program. The user password is also called a system password because it controls access to the entire system. Table 5.5 shows the security functions in a typical BIOS Setup.

Table 5.5: Typical Security Settings

Feature	Options	Description
Supervisor Password	—	Reports if a supervisor password is set.
User Password	—	Reports if a user password is set.
Set Supervisor Password1	—	Specifies the supervisor password.
User Access Level2	Full (default) No Access View Only Limited Only	Sets the user access rights to the BIOS Setup Utility. Full allows the user to change all fields except the supervisor password. No Access prevents user access to the BIOS Setup Utility. View allows the user to view but not change the BIOS Setup Utility fields. Limited allows the user to change some fields.
Set User Password1	—	Specifies the user password.
Clear User Password3	OK (default) Cancel	Cleans the user password.
Chassis Intrusion	Disabled (default) Log Log, notify once Log, notify until cleared	Disabled disables chassis intrusion; Log logs the intrusion in the event log. Log, notify once halts the system during POST, the user must press F4 to continue, the intrusion flag is cleared, and the event log is updated. Log, notify until cleared halts the system during POST and the user must enter the BIOS Setup Security menu and select Clear Chassis Intrusion Status to clear the chassis intrusion flag.

1. Passwords can be up to seven alphanumeric characters, including A-Z, a-z, and 1-9.
2. This feature is displayed only if a supervisor password has been set.
3. This feature is displayed only if a user password has been set.

Power Menu

Power management is defined as the capability of the system to automatically enter power-conserving modes during periods of inactivity. Two main classes of power management exist; the original standard called Advanced Power Management (APM) and the new type called Advanced Configuration and Power Interface (ACPI). Table 5.6 show the typical power settings found in a managed system.

Table 5.6: Typical Power Settings for a Managed System

Feature	Options	Description
ACPI	—	Selects a submenu that sets the ACPI power management options.
After Power Failure	Last State (default) Stay Off Power On	Specifies the mode of operation if an AC power loss occurs. Last State restores the computer to the power state it was in before the power loss. Stay Off keeps the computer powered off until the power button is pressed. Power On boots the computer when power is restored.
Wake on PCI PME	Stay Off (default) Power On	Specifies how the computer responds when system power is off and a PCI power management event occurs.

Boot Menu (Boot Sequence, Order)

The Boot menu is used for setting the boot features and the boot sequence. Table 5.7 shows the functions and settings available on a typical motherboard.

Table 5.7 Typical Boot Menu Settings

Feature	Options	Description
Silent Boot	Enabled (default) Disabled	Enabled displays the OEM graphic instead of POST messages. Disabled displays normal POST messages.
Add-On ROM Display Mode	Enabled (default) Disabled	Enables/disables the splash screen for add-in cards.
Intel Rapid BIOS Boot	Enabled (default) Disabled	Enables the computer to boot with or without running certain POST tests.
PXE Boot to LAN	Disabled (default) Enabled	Disables/enables PXE boot from LAN. When set to Enabled, you must reboot for the Intel Boot Agent device to be available in the boot Device menu.
USB Boot	Enabled (default) Disabled	Enables/disables booting from USB devices.
Boot Device Priority	—	Selects a submenu that specifies the boot sequence from the available types of boot devices.

Hard Disk Drives*	—	Selects a submenu that specifies the boot sequence from the available hard disk drives.
Removable Devices*	—	Selects a submenu that specifies the boot sequence from the available removable devices.
ATAPI CD-ROM Drives*	—	Selects a submenu that specifies the boot sequence from the available ATAPI CD-ROM drives.

**This feature is displayed only if this type of device is present.*

Exit Menu

The Exit menu is for exiting the Setup program, saving changes, and loading and saving defaults. Table 5.8 shows the typical selections found in most motherboard BIOSs.

Table 5.8: Typical Exit Menu Settings

Feature	Description
Exit Saving Changes	Exits and saves the changes in CMOS RAM.
Exit Discarding Changes	Exits without saving any changes made in the BIOS Setup.
Load Optimal Defaults	Loads the factory (optimal) default values for all the Setup options.
Load Custom Defaults	Loads the custom defaults for Setup options.
Save Custom Defaults	Saves the current values as custom defaults in CMOS RAM. Normally, the BIOS reads the saved Setup values from CMOS RAM. If this memory is corrupted, the BIOS reads the custom defaults. If no custom defaults are set, the BIOS reads the factory defaults from the flash ROM.
Discard Changes	Discards changes without exiting Setup. The option values present when the computer was turned on are used.